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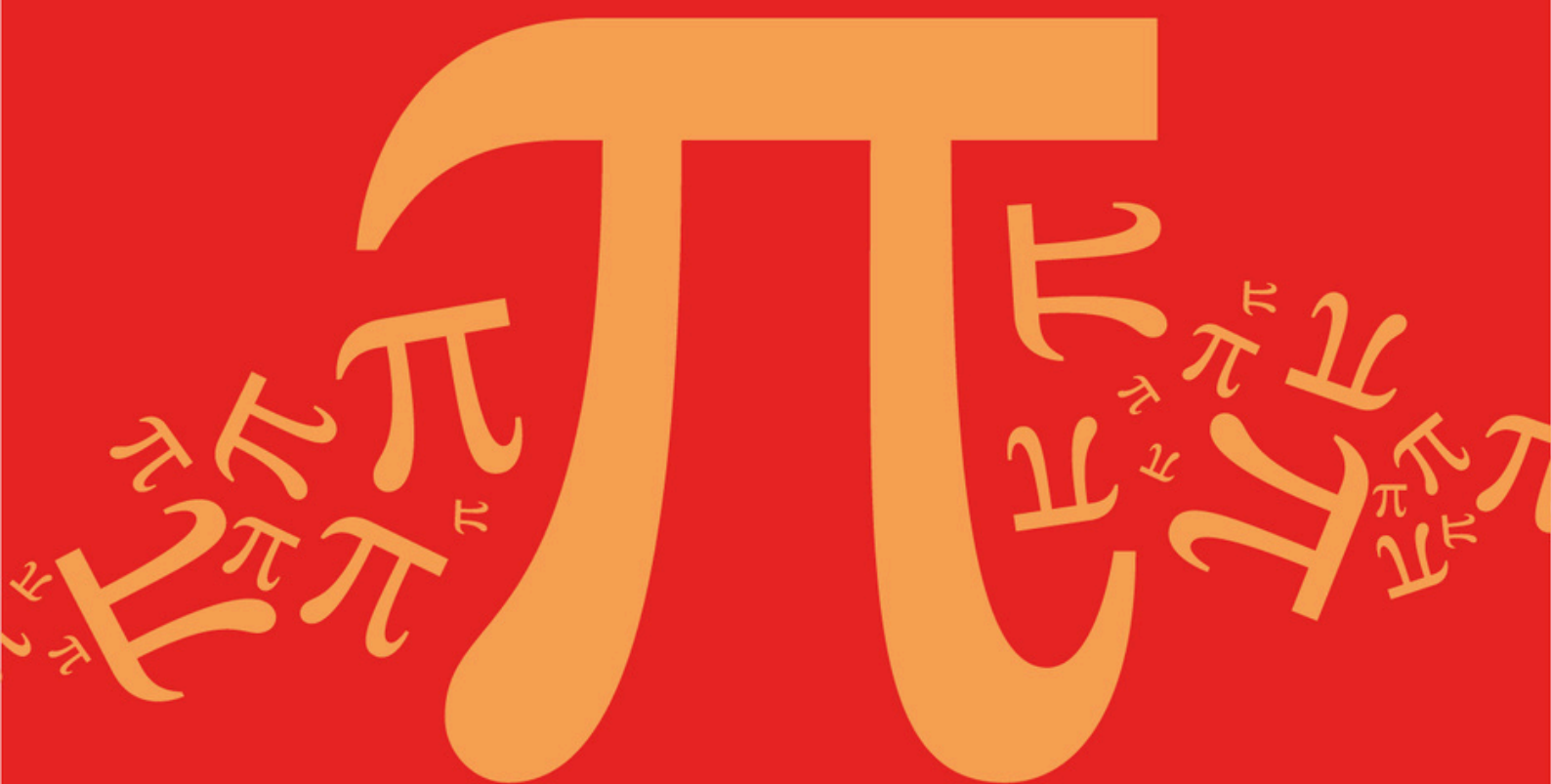
Detailed mark scheme

Suitable for all boards

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AQA A Level Further Mathematics

(7367)



Topic

D: Further Algebra and Functions

Sub topic

D12: Graphs of Rational Functions

Mark Scheme

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D12: Graphs of Rational Functions

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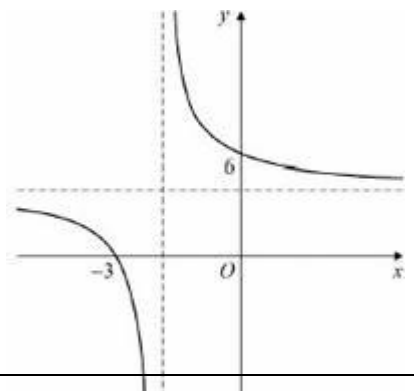
**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
(a) Writes any rational function with a horizontal asymptote of $y = 2$ or one vertical asymptote of $x = -1$ e.g. $y = \frac{ax+b}{x+1}$ or $y = \frac{2x+b}{x+c}$ or $y = \frac{ax^n+bx^{n-1}+cx^{n-2}+\dots}{dx^n+ex^{n-1}+fx^{n-2}+\dots}$ where $\frac{a}{d} = 2$ Accept any correct rearrangement of $y = f(x)$, where $f(x)$ is a function as described above.	AO3.1a	M1	$y = \frac{2x+m}{x+1}$
Obtains a fully correct answer.	AO1.1b	A1	but $x = -3$ when $y = 0$ $\therefore 0 = \frac{2 \times -3 + c}{-3 + 1}$ $-6 + c = 0$ $c = 6$ $\therefore y = \frac{2x+6}{x+1}$
			$(x+1)(y-2) = n$ but $x = -3$ when $y = 0$ $\therefore (-3+1)(0-2) = n$ $4 = n$ $\therefore (x+1)(y-2) = 4$ $y-2 = \frac{4}{x+1}$ $y = 2 + \frac{4}{x+1}$

Marking Instructions	AO	Marks	Typical Solution
(b) Sketches any rectangular hyperbola, or rational function, tending to the correct vertical and horizontal asymptotes included or implied.	AO1.1a	M1	
Sketches a correct graph, including the asymptotes. Accept the graph of their function if M1A1 scored in part (a). Accept un-ruled asymptotes – mark	AO1.1b	A1	

	intention.			
	Indicates correct axis-intercepts. Follow through their equation if their y-intercept matches their graph.	AO1.1b	A1F	
(c)	Forms an equation or inequality with $y = 5$ and their rational function.	AO1.1a	M1	$5 = \frac{2x+6}{x+1}$ $5(x+1) = 2x+6$ $5x+5 = 2x+6$ $3x = 1$ $x = \frac{1}{3}$ $x < -1, x \geq \frac{1}{3}$
	Obtains correct x-intercept with $y = 5$ Follow through their rational function from part (a)	AO1.1b	A1F	
	Deduces one correct region $x \geq \frac{1}{3}$ or $x < -1$ Condone $x \leq -1$ for this mark. Follow through their $\frac{1}{3}$ if greater than -1	AO2.2a	A1F	
	Deduces correct regions. Accept correct regions for their function if M1A1 scored in part (a).	AO2.2a	A1	

	$\frac{2x+6}{x+1} \leq 5$ $(2x+6)(x+1) \leq 5(x+1)^2$ $0 \leq (x+1)(5(x+1) - (2x+6))$ $0 \leq (x+1)(3x-1)$ $\begin{array}{c} + \quad \quad - \quad \quad + \\ \hline -1 \quad \quad \frac{1}{3} \end{array} \rightarrow x$ $x < -1, x \geq \frac{1}{3}$
	$\frac{2x+6}{x+1} \leq 5$ $\frac{2x+6}{x+1} - \frac{5(x+1)}{x+1} \leq 0$ $\frac{-3x+1}{x+1} \leq 0$ $\begin{array}{c} - \quad \quad + \quad \quad - \\ \hline -1 \quad \quad \frac{1}{3} \end{array} \rightarrow x$ $x < -1, x \geq \frac{1}{3}$

	For $x > -1$: $2x + 6 \leq 5(x + 1)$ $1 \leq 3x$ $x \geq \frac{1}{3}$ For $x < -1$: $2x + 6 \geq 5(x + 1)$ $1 \geq 3x$ $x < -1, x \geq \frac{1}{3}$
Total 9 marks	

Q2.

- (a) Asymptotes $x = 1$
 $x = 1$ OE

B1

$y = 1$
 $y = 1$ OE

B1

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- (b) $-4x + c = \frac{x}{x-1}$
Elimination of y PI by next line

M1

$$\begin{aligned}
 (-4x + c)(x - 1) &= x \\
 -4x^2 + cx + 4x - c &= x \\
 -4x^2 + cx + 3x - c &= 0 \\
 &\text{OE (denominators cleared)}
 \end{aligned}$$

A1

$$\begin{aligned}
 4x^2 - (c + 3)x + c &= 0 \\
 &\text{CSO AG No incorrect algebraic expressions etc}
 \end{aligned}$$

- (c) (i) Discriminant is $(c + 3)^2 - 4(4c)$
 OE

For tangency $c^2 - 10c + 9 = 0$

Forming a quadratic eqn in c after equating discriminant to zero

M1

$$(c - 9)(c - 1) = 0 \Rightarrow c = 1, c = 9$$

Correct values 1, 9 for c .

A1

3

(ii) $\underline{c = 1}$: $4x^2 - 4x + 1 = 0$

Substitutes at least one of c 's values for c from (c)(i)

either into the given quadratic in (b) OE or into $\frac{x}{x-1}$

M1

$$\underline{c = 9}$$
: $4x^2 - 12x + 9 = 0$

$$4x^2 - 4x + 1 = 0 \Rightarrow x = 1/2 \quad (= 0.5)$$

No other root from quadratic

A1

$$4x^2 - 12x + 9 = 0 \Rightarrow x = 3/2 \quad (= 1.5)$$

No other root from quadratic

A1

When $x = 1/2$, $y = -1$; when $x = 3/2$, $y = 3$

$$\left(\frac{1}{2}, -1\right) \quad \left(\frac{3}{2}, 3\right)$$

Accept in either format

A1

4

[12]

Q3.

- (a) (i) Asymptotes $x = -2$, $x = 2$, $y = 0$

B1 $\times$ 3

3

- (ii) Middle branch generally correct
Allow if max pt not in right place

B1

Other branches generally correct

B1

All branches approaching asymptotes

B1

Intersection at $\left(0, -\frac{1}{4}\right)$ indicated

Asymptotes must be shown correctly on diagram or elsewhere; B0 if any other intersections are shown

B1

4

(b) $y = -2$ when $x = \pm \sqrt{3.5}$

Allow NMS

B1

Sol'n $-2 < x < -\sqrt{3.5}, \sqrt{3.5} < x < 2$

*Condone dec approx'n for $\sqrt{3.5}$;
B1 if $\leq$ used instead of $<$*

B2,1

3

[10]

Q4.

- (a) (i) Asymptotes $x = 3$ and $y = 0$
may appear on graph

B1,B1

2

- (ii) Complete graph with correct shape

B1

Coordinates $\left(0, -\frac{1}{3}\right)$ shown

B1

2

- (iii) Correct line, $(0, -5)$ and $(2.5, 0)$ shown

B1

1

(b) (i) $2x^2 - 11x + 14 = 0$

B1

$x = 2$ or $x = 3.5$

M1 for valid method for quadratic

M1A1

3

(ii) $2 < x < 3, x > 3.5$

*B1 for partially correct solution;
ft incorrect roots of quadratic
(one above 3, one below 3)*

B2,1F

2

[10]

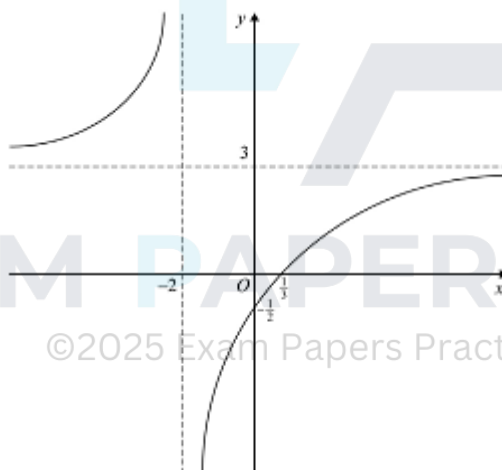
Q5.

(a) Asymptotes $x = -2, y = 3$

B1,B1

2

(b)



Curve approaching asymptotes

B1

Passing through $\left(\frac{1}{3}, 0\right)$ and $\left(0, -\frac{1}{2}\right)$

B1,B1

*Both branches generally correct
B1 if two branches shown*

B1,B1

5

(c) Solution set is $x > \frac{1}{3}$

*B1 for good attempt;
ft wrong point of intersection*

B2,1F

2

[9]

Q6.

- (a) Asymptotes $x = 1$, $y = 6$

B1B1

2

- (b) Curve (correct general shape)

M1

Curve passing through origin

A1

Both branches approaching $x = 1$

A1

Both branches approaching $y = 6$

A1

SC Only one branch:

B1 for origin

B1 for approaching both asymptotes

(Max 2/4)

4

- (c) Correct method

M1

Critical values ± 1

From graph or calculation

B1B1

Solution set $-1 < x < 1$

ft one error in CVs; NMS

4/4 after a good graph

A1ft

4

[10]

Examiner reports

Q1.

In part (a), most students could write a function with at least one of the required asymptotes. Those students who found a function which satisfied all of the necessary criteria generally went for the straightforward rectangular hyperbola.

Most students were able to draw a suitable graph in part (b), although some sketches failed to show a correct approach to the asymptotes.

In part (c), many students found a correct intersection with $y = 5$, but a significant number failed to successfully progress further.

Q2.

Most candidates were able to state the correct equations of the two asymptotes. Answers to part (a) which were not equations did not score the marks.

The method required to answer part (b) was well understood, but candidates who made sign errors, omitted brackets and still reached the printed answer were not awarded full credit.

Most candidates realised that the discriminant of the quadratic was to be used to answer part (c)(i) but a number of candidates failed to state that its value had to be 0 for the line to be a tangent. Some used an inequality and then, in finding the critical values, just stated the two values of c without any reference back. Some errors were seen in finding the discriminant, normally due to the negative sign and squaring $-(c + 3)$ incorrectly to get $-c^2 - 6c - 9$. This, and similar errors, led to horrendous quadratics and candidates would have been better advised to check earlier working rather than try to solve them. There were, however, many fully correct solutions seen to this final question on the paper.

Q3.

The asymptotes asked for in part (a)(i) were usually given correctly, and there was a high standard of curve sketching in part (a)(ii). In part (b), only a small proportion of candidates gave the full correct solution of the inequality. As well as looking at the sketch-graph, it was necessary to do a calculation here, and many candidates earned one mark out of three by finding the two correct points of intersection of the curve with the line $y = -2$.

Q4.

The first eight marks out of the ten available in this question were gained without much difficulty by the majority of candidates, apart from some careless errors such as omitting to indicate the coordinates asked for on the sketch.

By contrast the inequality in part (b)(ii) was badly answered. Few candidates seemed to think of reading off the answers from the graph, the majority preferring an algebraic approach, which if done properly would have been worth much more than the two marks on offer. The algebraic method usually failed at the first step with an illegitimate multiplication of both sides of the inequality by $x - 3$. Some candidates multiplied by $(x - 3)^2$ but could not cope with the resulting cubic expression.

Q5.

Many candidates scored well on parts (a) and (b) of this question, but relatively few candidates made a good attempt at part (c). In part (a) the asymptotes were usually found correctly, as were the coordinates of the required points of intersection in part (b). The graph in part (b) was usually recognisable as a hyperbola, or at least as one branch of a hyperbola, the other branch not being seen. In part (c) many candidates resorted to algebraic methods for solving inequalities rather than simply reading off the solution set from the graph. These algebraic methods, more often than not, were spurious.

Q6.

Parts (a) and (b) of this question were generally well answered, though some otherwise good graphs did not pass through the origin as required. The inequality in part (c) was not tackled at all well. Few candidates seemed to realise that a good graph allowed them to read off the answer, the only further detail required being to ascertain the value of x for which $y = 3$. A distressingly large number of candidates simply multiplied both sides of the inequality by the denominator ($x - 1$). Those who adopted sounder methods often arrived at the statement $x^2 < 1$, but then showed a lack of mathematical awareness by drawing a meaningless conclusion such as $x < \pm 1$.